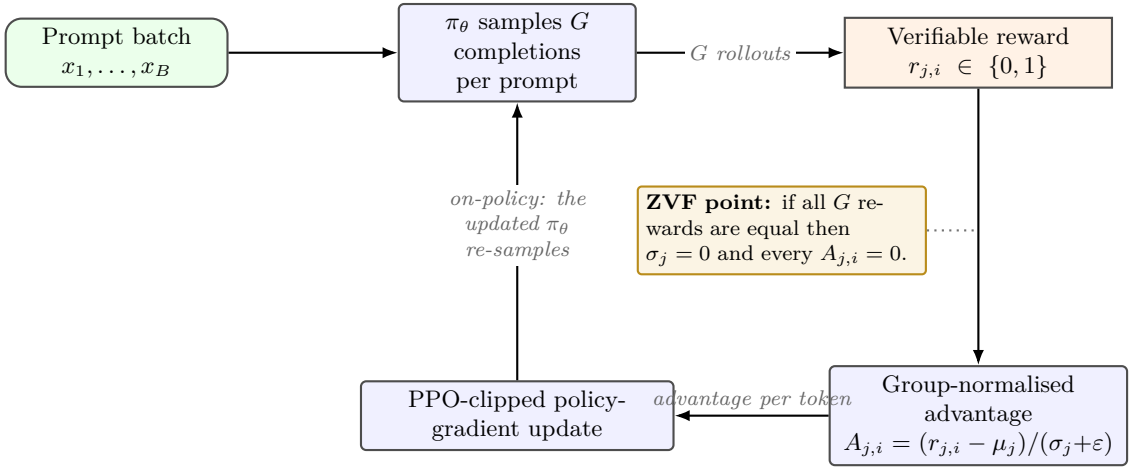


The ZVF point, in detail

Degenerate group: $\sigma_j = 0$

1	1	1	1	1	1	1	1

$\mu_j = 1.00 \qquad \sigma_j = 0.00$

$\Rightarrow A_{j,i} = (1 - 1)/(0 + \varepsilon) = 0$ for every i

The surrogate term vanishes, so the group contributes *no* reward-relative gradient. If enabled, KL or auxiliary-loss gradients may remain; their presence does not guarantee a nonzero optimizer update. The same reward-relative zero applies when all rewards are 0.

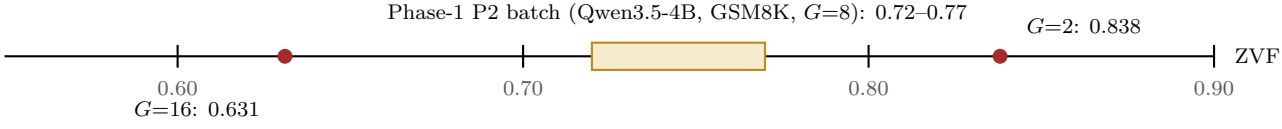
Informative group: $\sigma_j > 0$

1	0	1	1	0	0	1	0

$\mu_j = 0.50 \qquad \sigma_j \approx 0.53$

$\Rightarrow A_{j,i} \approx \pm 0.94$: oppositely signed advantages

Correct and incorrect completions have oppositely signed reward-relative advantages. Clipping, model derivatives and aggregation determine the resulting gradient; nonzero advantages alone do not guarantee a nonzero optimiser update.



as the group grows (G:2 $\rightarrow$ 16) the ZVF falls by 0.207; sweep: Qwen2.5-0.5B, two-digit addition, 3 seeds; in the P2 regime the collapse above is reached for roughly three quarters of groups